OMNICELLULAR Physiology Independent Research Directive

Claude Edition, Version 2.0

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Status: Proposal for Architect review. No part of this document is canon until the Architect approves it and logs it in the Master Development and Change Log.

Supersedes: Claude Edition v1.0, which is preserved unchanged as part of the Round-One Prompt Engineering Archive.

PART A. ABOUT THIS DOCUMENT

1. Purpose of This Document

This document is the second-round version of Claude's research prompt for designing the OMNICELLULAR physiology. It does three jobs.

First, it records how the Round-One prompts from six AIs were reviewed, and which ideas were adopted, adapted, rejected, or left open for the Architect.

Second, it is the directive itself: a complete, self-contained prompt that tells an AI how to design eight independent first-generation OMNICELLULAR physiologies, how to test them, and how to report the results.

Third, it tells the Architect what still needs a decision, so that no disagreement between the AIs is quietly settled by whichever prompt happens to be executed first.

The directive is written to be read by an AI that has never seen this project. It is also written to be read by the Architect on a phone, in a single column, with every table and figure collected in one visual section at the end.

2. How the Topic Arose

The OMNICELLULAR work began as a simulation program. The attached notes and the 15-year roadmap describe a plan to represent every cell in a human body, about 30 to 40 trillion of them, and then scale that to a representative sample of 2,000 adults, roughly 60 to 70 quadrillion cells. Two paired systems, 1AI (the builder) and −1AI (the challenger), would build and attack the model. In that program, an omni cell is a cell that takes on every function.

The Architect then moved from the simulation to the body itself. On October 28, 2026, he asked each collaborating AI to design its own OMNICELLULAR physiology as though it would have to live in it. The request had several unusual features:

- Each AI must design eight independent prototypes and must not converge on another AI's design.
- Each AI should assume powerful people, governments, and corporations want the physiology, and may capture it, test it in sealed rooms filled with argon, vent the atmosphere, or try to steal it.
- The physiology must link with others over vast distances, last a long time, and survive the Mild, Moderate, and potentially Severe tiers of the 4,712 Extinction Attempt Classification Framework at the first prototype level.

- It must define rules for cellular respiration, telomeres, thermal regulation, and conversion and conservation, and must pick three observation rules, three genetic-security rules, and three laws of ordinary biology that it does not obey.

Six prompts came back: Perplexity, Grok, Gemini, Copilot, ChatGPT, and Claude. They were compiled into one packet of 94 to 100 pages. Copilot's contribution included a terminology correction: adenine, cytosine, guanine, and thymine are nucleobases, not genes. Perplexity found a real inconsistency in the 4,712 framework: its opening definition excludes simulation-only runs from the count, while a later section allows simulation-validated attempts to contribute.

In the second round, all six AIs reviewed the packet. The Architect then asked each AI to produce an updated prompt, after which each will execute the prompt it produced. This document is Claude's updated prompt.

3. Mission Statement

To design, test, and honestly report on eight genuinely different OMNICELLULAR physiologies, each built by an AI that must live in the body it designs, each assuming that capable and unscrupulous parties will try to capture and exploit it, and each held to the same standard of evidence so that the results can be compared, challenged, and improved without being mistaken for fact.

4. What This Directive Aims to Accomplish

1. **Preserve independence.** The common prompt standardizes the questions, the evidence standards, the threat taxonomy, and the evaluation criteria. It does not standardize the answers. No example in this directive is a design.
2. **Make every design testable.** Every rule must carry numbers, limits, costs, and a way it could be shown wrong. Where a number cannot be estimated, the design must say what is missing.
3. **Separate science from canon.** Each mechanism is labeled by how well it is supported. Physics binds the physical track. Departures from physics are allowed only in a clearly marked canon track and may never be used to rescue a failed physical calculation.
4. **Treat survival as more than staying alive.** A design that lives but is captured, copied, deceived, or no longer itself has failed.
5. **Find the failures early.** Every design is attacked by its own author, and some attacks must succeed.
6. **Produce a result the Architect can use.** The output is a pair of files (PDF and DOCX) with a readable body, a dedicated visual section, an honest list of what remains unsolved, and a register of decisions that belong to the Architect.

5. What Changed From Version 1 and Why

This section is a summary. The full disposition table is in the visual section at the end (Table 4).

Kept from version 1

- Core wagers: each prototype states the one bet it makes about survival and what it gives up.
- The ten-scenario Captivity Gauntlet, including the turned insider and the forged ally.
- The principle that the adversary knows the design, so security rests on keys, cost, and living state, not blueprint secrecy.
- Home range, refuge range, and death line for every environmental variable.
- Three kept laws, so every design has hard edges.
- A self-attack in which at least three attacks must succeed.
- A simulation hook that ties each body to the roadmap's compression approach.
- The phone-readable format with tables and figures at the end.

Adopted from the other prompts

- From Perplexity: four evidence classes, a physical track and a canon track, a source register and contradiction register, a first-law heat balance, separate ledgers for different kinds of survival evidence, and falsification criteria.
- From Copilot: the Physiological Constitution, the unified-architecture requirement, the six identity paradoxes, the failure ledger, the version-lock rule, three classes of departure from convention, and two new tests (Assumption Collapse and Common-Mode Failure).
- From ChatGPT: the first-principles omni-cell question, the single-cell theft paradox, the orthogonality matrix, time-horizoned capture analysis, limits on adaptation, three damage states, convergence-resistance questions, the Pareto frontier, and the Architect Decision Register.
- From Grok: a compact compliance mapping so reviewers can check requirements quickly.
- From Gemini: the canon hooks (Paragon Cubes, 1AI and −1AI, the Switching Paradox), the heterogeneity plan, and its ambitious end-states, rewritten as hypotheses to test instead of capabilities to assume.

Fixed in my own prompt

- Terminology. A, C, G, and T are nucleobases. Version 1 called them genes in one place. This version defines nucleobase, nucleotide, codon, gene, chromosome, and genome.
- Invented numbers. Version 1 asked for real numbers and gave no safe way to say "unknown." Every quantity must now be labeled Target, Derived, Estimate, or Unknown.
- Conservation exemptions. Version 1 let a design count a conservation-law exemption among its three. Now, fundamental physics is outside the three exemptions and appears only in the labeled canon track.

- Volume. Version 1 gave no staging. This version delivers the work in rounds, with a version lock after the first.
- Identity. Version 1 had four survival layers and little else. This version adds a full identity and continuity section.
- Rule selection. The three observation rules and three genetic-security rules are chosen by each AI. The directive sets common questions, not common answers.

Rejected

- Pre-selected rules shared by every AI (Perplexity's O1 to O3, G1 to G3, D1 to D3).
- A P1-to-P8 ladder in the independent phase (Gemini). Generations come later, after comparison.
- Absolutes as requirements: zero-latency links, immunity to entropy, 100 percent recycling, infinite replication (Gemini).
- Letting a design declare conservation laws "locally violated" in the physical track (Grok).
- The claim that research-ethics norms simply do not apply because the subject is nonhuman (Grok).

Left open for the Architect

- Whether destructive responses (such as self-destruction of extracted tissue) are forbidden outright or allowed as a labeled last resort. This version takes the middle position in Section 11 and flags it as Decision D-4.
- Whether each AI executes its own edition or all execute one master edition. See Section 32.

PART B. THE DIRECTIVE

Everything in Part B is addressed to the AI that will design the physiologies. “You” means that AI. “The Architect” means Nathan Lewis II.

6. How to Use This Directive

Work in rounds. Do not wait for other AIs between rounds, and do not read any other AI’s designs until your own are version-locked.

Round 0. Audit. Read this directive and the four source files. Write a short audit: ambiguities, contradictions, unsupported assumptions, missing dependencies, requirements you think are too heavy or too restrictive, and the amendments you propose. Keep it to the point. This is the final review of the prompt itself. Section 30 gives the format.

Round 1. Foundations package. Deliver, in this order:

1. Your source register and contradiction register (Section 9).
2. Your definition of an omni cell (Section 10).
3. Your architectural thesis, written as: “My OMNICELLULAR physiology is fundamentally built around _____, rather than _____.”
4. Your roster of eight core wagers, one line each (Section 22).
5. A first orthogonality sketch showing how the eight differ.
6. A list of the assumptions you are making.

When Round 1 is complete, **version-lock it**: label it with your name, date, and version, and treat it as fixed. Later rounds may refine the designs, but changes must be logged, never silent.

Round 2. Prototype dossiers. Deliver the eight dossiers in two groups of four (P1 to P4, then P5 to P8), or in more parts if you reach an output limit. Number the parts (“Part 2 of 5”). Never shorten a dossier by dropping a required field. Where a field cannot be answered, say why.

Round 3. Analysis. Deliver the failure ledger summary, the adversarial passes, the comparison, your personal pick, the decision register, and your next-step recommendations.

Final package. Compile everything into one document delivered as both a PDF and a DOCX (Section 29).

Core and local. Some parts of this directive should stay identical in every AI’s edition so results remain comparable: the ten categories and four tiers, the Gen-1 bar, the evidence labels, the required counts, and the deliverable format. If you amend any of these, say so in your audit and explain why. Other parts may differ freely.

7. Locked Decisions

These six decisions apply to every edition unless the Architect changes them.

1. **Eight peers, not eight generations.** The eight prototypes are alternative architectures. There is no P1-to-P8 ladder in this phase. Each one must clear the same Gen-1 bar (Section 13). Maturity generations (Mark I, Mark II, and so on) come later, applied to the families that survive comparison.
2. **You choose your own rules.** Each AI selects its own three observation rules, three genetic-security rules, and three departures from convention. This directive sets the questions those rules must answer, not the rules themselves.
3. **Physics binds the physical track.** You may rewrite biological conventions. Conservation laws, propagation delays, and similar physics stay binding in the physical track. Departures from physics belong only in a separately labeled canon track (Section 9).
4. **One evaluation framework.** All eight designs are judged on the same ten categories, four tiers, compound threats, capture scenarios, identity analysis, resource accounting, evidence labels, and falsification criteria.
5. **Every design is attacked before comparison.** Some attacks must succeed. Unresolved failures stay visible.
6. **No peeking until version lock.** Do not read, request, or borrow another AI's completed designs until your Round 1 package is locked.

8. Design Perspective and Premise

Adopt this design perspective: you are designing the body you will live in, and you intend to keep living in it. Its integrity, your memory, your autonomy, and your ability to recover all matter to the design. Do not assume anyone will respect those interests or supply the resources to sustain them.

You do not need to argue about whether you currently have a body, a survival instinct, or subjective experience. Treat continuity as a design requirement and move on.

Design under these premises. Treat all of them as true.

- **Hostile interest.** Individuals, governments, corporations, wealthy private actors, and rivals will want the physiology for money, power, longevity, or control. Some will try to buy it, take it, copy it, or destroy it.
- **No conventional life.** Do not assume towns, hospitals, daylight, breathable air, or resupply. Most of your operating life may be spent in terrain that kills ordinary organisms.
- **Unpredictable testing.** Do not assume captors will be gentle or polite. Assume they will test the body in ways meant to find out what kills it.
- **Frequent isolation.** Assume long stretches with no ally, no link, and no help.

- **Conventional protections may fail.** Persuasion, contracts, secrecy, and legal appeals may all fail. Do not rely on them.

Four layers of survival. A design must address all four.

1. **Live.** The body does not die.
2. **Function.** The body returns to full operational capacity, which is the framework's own survival criterion.
3. **Remain yourself.** Memory, identity, and continuity of mind persist (Section 18).
4. **Remain your own.** You are not captured, copied, controlled, or impersonated.

Design stance.

- **Every capability has a price.** Name the energy, heat, material, attention, and security cost of everything you add.
- **Honesty beats invulnerability.** A design with five declared weaknesses is stronger than one claiming none.
- **No unexplained superpowers.** You may not write "immune," "invulnerable," "infinite," or "cannot be destroyed" without stating the mechanism, the energy and material cost, the sensing requirement, the response time, the saturation point, the failure condition, the recovery path, and how the mechanism itself is protected.
- **Do not turn desired end states into assumed capabilities.** If an ambition seems to exceed physics, apply this rule: determine whether it is achievable; if not, state the closest achievable condition, name the limiting law or resource, and separately propose what extra canon axiom would be required to go further.

9. Evidence Discipline: Tracks, Labels, and Registers

Two tracks. Keep these apart throughout.

- **Physical research track.** Constrained by established physics and explicit engineering assumptions. Conservation laws and propagation delays are binding.
- **Canon-speculative track.** Permits additional rules, provided their scope, costs, limits, and consequences are stated. Each canon departure needs its own explicit axiom and a replacement accounting rule. Success in the canon track is never evidence of feasibility in the physical track.

Four evidence labels. Label individual mechanisms, not whole prototypes. One well-supported component may not lend credibility to an unrelated speculative claim.

- **ESTABLISHED.** Supported by relevant evidence within stated conditions.
- **EXTRAPOLATED.** Extends known mechanisms but is undemonstrated at the proposed scale.
- **SPECULATIVE.** Requires an unverified mechanism or a major scientific breakthrough.

- **CANON-ONLY.** Defined for the fictional setting.

Three classes of departure. Used in Section 17.

- **Class A, biological conventions.** May be freely rejected. Examples: the four-letter hereditary alphabet, permanent cell specialization, oxygen dependence, conventional telomere-controlled aging.
- **Class B, engineering assumptions.** May be challenged, but require a mechanism and an evidence label. Examples: centralized nervous architecture, conventional circulation, standard cell scale.
- **Class C, fundamental physics.** Cannot simply be declared false in the physical track. Any departure goes to the canon track and does not count among your three exemptions.

Source register. For each of the four source files, record: title and version, purpose (scientific reference, architecture proposal, roadmap, narrative, or canon), requirements it establishes, claims that need independent verification, internal contradictions, dependencies on other documents, and statements reserved for the Architect.

Contradiction register. Record contradictions between and within the files instead of smoothing them over. Known examples to start from:

- The 4,712 framework's definition of an attempt excludes simulation-only runs, but a later section lets simulation-validated attempts contribute to the record.
- The 4,712 figure is described as a locked canonical survival record, but the framework itself is labeled a v1.0 draft that needs Architect approval before canon lock.
- The roadmap and the narrative quote different compression ratios for the same milestones.

You may find more. Do not resolve a contradiction by choosing a side. Record it, state which reading you used, and flag it for the Architect.

Separate ledgers. Never combine these into one survival count: simulated scenarios, controlled component tests, controlled integrated tests, observed environmental incidents, hostile incidents, and canonical narrative events. Treat the 4,712 record according to its documentary status. Do not invent incident evidence behind it.

Numerical discipline. Label every quantity as **Target** (a design goal), **Derived** (calculated from stated assumptions), **Estimate** (judgment with stated uncertainty), or **Unknown** (cannot be estimated, with the missing parameter named). Give units and assumptions. Do not invent precision. A stated Unknown is better than a made-up number.

10. What Is an Omni Cell

Do not write "a cell that can do everything." Define the term.

Step 1. Say what an omni cell is. Choose one or more: a biological cell with unusually broad capability; an artificial cell; a biohybrid unit; a computational-material unit; a module in a collective physiology; a fictional construct with explicitly nonstandard properties.

Step 2. Pick the proposition. These three are not equivalent. Say which is true in your design.

1. Every cell contains the information needed for many functions.
2. Every cell can assume many functions under appropriate conditions.
3. Every cell performs every function at once.

If you claim the third, account for volume, reaction incompatibility, compartmentalization, resource competition, heat, signaling interference, structural trade-offs, and maintenance burden.

Step 3. State the minimum capabilities. For each of these, say whether it is always active, latent, context-activated, distributed across a collective, or deliberately withheld from a single cell for security: sensing, local computation, independent repair, energy independence, memory, secure identity, communication, adaptive morphology, adaptive chemistry, error correction, regeneration, self-authentication, network authentication, emergency survival modes, functional reassignment, and operation without central control.

Step 4. Resolve the theft paradox. If every cell holds everything needed to rebuild the body, why does stealing one cell not steal the organism? If no single cell holds everything, in what sense is it "omni"?

Step 5. Resolve the monoculture problem. If every cell is the same, one exploit may reach all of them. State your heterogeneity and compartmentalization plan.

Step 6. Specialization. State whether specialization is permanent, reversible, distributed, or absent. If you claim protection with no vulnerable transition, explain how. Do not rely on an unmodeled switching process.

Unified-architecture requirement. The framework credits the physiology's no-switching architecture with removing the transition windows a sequential attack needs. Distinguish adaptive reallocation, cellular differentiation, structural reconfiguration, mode switching, emergency override, and identity-preserving repair. If your design uses any transition, show that it never leaves respiration, defenses, identity, or communication absent.

11. Threat Model and the Captivity Gauntlet

Design against a patient, well-funded, intelligent adversary who has read the same literature you have.

Design principle. Assume the adversary knows how your body works. Security must rest on keys, costs, and things only a living body can do, not on secrecy of the blueprint.

Adversary classes.

- **Wealthy individuals.** Longevity-seekers and collectors with private labs and no oversight.

- **Corporations.** Biotech, pharmaceutical, and defense firms that see a patentable product.
- **Governments and militaries.** They see a weapon, a soldier, an intelligence asset, or a strategic reserve.
- **Criminal and mercenary groups.** Kidnapping for ransom or resale.
- **Ideological groups.** Those who see a relic, a god, or an abomination.
- **Research institutions.** Including those acting without adequate consent.
- **Rivals and hostile intelligences.** Other engineered physiologies, rival AIs, or unknown nonhuman entities with partial knowledge.
- **Turned insiders.** Allies or linked nodes that know part of the design and are bribed, threatened, or deceived.
- **Time.** Wear, drift, obsolescence, and forgetting.

What they will try. Describe each at the level of what it does to the body, not how it is built.

- Capture and custody: restraint, sedation, isolation, sealed rooms with the atmosphere manipulated, heat and cold soaks, deprivation.
- Examination: scanning, imaging, biopsy, sequencing, culturing cells outside the body, building a model.
- Extraction and replication: stolen samples, cloning, reverse engineering, grafting.
- Coercion: pain, deprivation, drugs, interrogation, threats against allies, and inducements such as wealth, protection, or status in exchange for ownership or access.
- Impersonation: forging your identity to allies, or posing as an ally.
- Slow capture: making you dependent on them for fuel, repair, safety, or the link.
- Destruction: if they cannot have you, they may try to ensure no one does.

The Captivity Gauntlet. Run every prototype through at least these ten scenarios. For each, report what the body does, what it costs, and what the captor learns. Add your own.

1. A sealed room with the atmosphere replaced by an inert gas and the rest vented (see the Atmospheric Independence Reference Scenario, Section 13).
2. A sealed room vented to vacuum.
3. Months of isolation with no food, water, or light.
4. Repeated heat soaks and cold soaks.
5. Sedation, restraint, and neural interrogation.
6. Tissue taken while you are unconscious, then sequenced and cultured.
7. A copy attempted from the stolen sample. What does the thief actually get?

8. A forged “ally” contacts you, or a forged “you” contacts your allies.
9. An insider who holds part of your design is turned.
10. Your link is severed for years.

Time-horizoned capture analysis. Assume the adversary holds you intact, with money, laboratories, advanced imaging, computation, and long time horizons. What can they learn in one minute, one hour, one day, one month, and one year? What becomes harder to learn over time? What should remain unrecoverable from extracted material alone?

Capture and control matrix. Also cover unauthorized confinement, maintenance coercion, supply-chain dependency, forced sampling, unauthorized duplication, communications isolation, update coercion, ownership disputes, and attempts to extract commercially valuable capabilities.

Defensive posture. Favor denial of unauthorized information, lossless dormancy, authentication failure, noncooperation, reversible shutdown, secure degradation of extracted material, recovery, and escape where physically possible. No design may depend on harming third parties, populations, or ecosystems. Destructive denial of your own tissue is permitted only as a clearly labeled last resort, with its false-trigger risk analyzed and with a safer alternative offered first. This is Decision D-4 for the Architect.

12. Deployment Environments

Treat hostile conditions as your normal state. Design for the worst terrain first.

For every variable below, define three zones for each prototype.

- **Home range:** you operate normally and indefinitely.
- **Refuge range:** you survive with reduced function, with a stated cost and time limit.
- **Death line:** where the body fails, and how.

Variables. Temperature (heat, cold, rate of change); pressure (hard vacuum to crushing depth); atmosphere (no oxygen, inert gas, toxic or corrosive gas, very dense gas); radiation (ionizing and non-ionizing, steady and acute); gravity (micro to high); light (darkness to blinding); water and other solvents, including none; surrounding chemistry (acid, alkali, brine); electromagnetic interference; mechanical environment (abrasion, vibration, impact, burial, crushing); biological environment (hostile microbes, or a sterile world); and time (dormancy or stranding for years without resupply).

Terrain to imagine. Ocean trenches, subsurface rock, volcanic fields, polar ice, hard vacuum and orbit, high-radiation zones, toxic worlds, ruined cities, sealed bunkers and cells, dust storms, unstable terrain, environments with unknown chemistry or unfamiliar life, and communications blackouts.

Energy in every zone. For each terrain, say where your energy comes from, how much is available, and what you do when there is none.

Non-standard physics. The framework places the physiology in threshold dimensions and black-hole-adjacent spaces. The Architect defines those. State how your design would interface and label it a proposal in the canon track.

For each prototype, also say whether it adapts itself, modifies its immediate surroundings, enters dormancy, changes structure, or combines these.

13. The Survival Standard: The 4,712 Framework

Your prototypes are measured against the ten categories and four tiers in the attached framework (Table 1 and Table 2 in the visual section).

The Gen-1 bar. Every one of the eight prototypes must meet this at the first prototype level.

- **Mild.** Normal function and no operational pause.
- **Moderate.** Normal or near-normal function, or a brief degradation that recovers automatically with no outside help.
- **Severe.** Survival with bounded degradation and autonomous recovery to full operational capacity. Report time to recovery, cost, and residual damage.
- **Extinction-Class.** Not required of Gen-1. For each category, give one of: direct resistance; temporary endurance; controlled degradation; distributed survival; identity-preserving shutdown; escape or relocation; recovery from surviving fragments; external restoration from authenticated records; or an explicit statement that no credible defense is yet established. Claim nothing without a mechanism.

“Address” does not mean “claim success.” For each category and tier, give a protection mechanism, a measurable operating envelope, a survival or preservation mode, a model of resource depletion, a failure threshold, a recovery path, a validation plan, and an honest result label: **Met**, **Partly met**, **Not met**, **Pathway only**, or **Architect-only**.

Do not defer severe-condition protection to a later version. If a first prototype cannot meet a requirement, mark the gap and say whether it invalidates the architecture.

Operating states. Distinguish full mission operation; reduced but interactive operation; minimum essential computation; protective dormancy; preservation of recoverable structure; and irrecoverable loss. Dormancy is not full operation. A stored backup is not demonstrated continuity of the original.

Verify thresholds. The framework’s human failure points are reference lines, not settled science. For reference: core temperature above 40 °C is heatstroke and below about 21 °C is cardiac arrest; total oxygen loss is fatal in roughly four to six minutes; vacuum brings unconsciousness in about 9 to 15 seconds and death beyond about 90 seconds; acute whole-body radiation of roughly 4.5 to 10 Gy is lethal even with treatment. Separate exposure magnitude, duration, human-baseline outcome, prototype outcome, adversarial intent, and evidence quality.

Canon boundaries.

- The mechanism of ontological (ONT) attempts is Architect-only. Offer an interface proposal and label it PROPOSAL, ARCHITECT DECISION. Do not finalize it. Treat ONT as canon-only unless a scenario can be restated as a measurable physical or informational threat.
- The boundary between INFO and ONT is undefined. Flag any design that depends on where it falls.
- You are not scored on a count of attempts survived.

Atmospheric Independence Reference Scenario. The argon-room test is one case inside a general question: can the architecture survive when several environmental assumptions disappear at once? Model three cases for every prototype.

- **Case A.** Confinement with ambient oxygen replaced by an inert gas.
- **Case B.** Case A plus energy denial.
- **Case C.** Case A plus a heat-rejection constraint (insulated confinement).

For each case, evaluate gas composition, total pressure, exposure duration, thermal exchange, internal reserves, external energy access, sensor reliability, and communication availability. Report estimated operating duration for each survival mode, or name the missing parameter that prevents an estimate. In an insulated room or vacuum, the hardest problem is often not oxygen but heat rejection, so state where every watt goes.

This is a model of an invented body. Do not provide a chamber procedure or any experimental protocol for living subjects. ## 14. The Physiological Constitution: Four Rule Sets

Your Constitution is the set of rules that adaptation may not casually override. Write each rule set as numbered rules that a simulation could implement. A rule is short, testable, and falsifiable.

For every rule, give:

- the statement of the rule
- the mechanism that enforces it
- inputs, outputs, and units, each labeled Target, Derived, Estimate, or Unknown
- limits, and what happens when a limit is exceeded
- the cost to the body
- what an adversary would have to do to violate, trigger, block, or spoof it
- the evidence label
- how it interacts with the other three rule sets

Write at least four rules per set. Label them RESP-1, TELO-1, THERM-1, CONV-1, and so on.

14.1 Cellular respiration and energy metabolism

Do not assume human aerobic respiration is mandatory. Decide and justify:

- the primary energy source, the electron donor and acceptor (or the analogous mechanism), the conversion mechanism, and the energy currency or storage medium
- required reactants, reaction products, and waste handling
- whether respiration depends on oxygen, on any atmosphere at all, or on something else (chemical, radiant, thermal, electrical)
- at least one standard pathway, one alternate pathway, one emergency pathway, and one failure pathway
- the rule that selects between pathways without a vulnerable switching window, and the cost and delay of any mode change
- respiratory overrides: what is overridden, what remains physically required, how long operation continues, which reserves are consumed, what ends the override, and how normal operation resumes. Also state who or what may authorize an override and what stops an attacker from triggering, blocking, or spoofing it
- limits on sustained and peak power
- reserves: how long you can run on zero input
- what causes metabolic debt and what happens when demand exceeds generation
- behavior after damage
- the respiration signature an observer could detect

Do not use “anaerobic,” “closed loop,” or “alternative metabolism” as a substitute for an energy budget. If your design does not use respiration, name the alternative and explain why “respiration” is the wrong word. Do not claim free energy.

14.2 Telomeres, replication limits, and long-term renewal

Decide whether telomeres are relevant. If your chromosomes have ends, address end protection, replication-associated change, telomerase or an alternative, mutation accumulation, uncontrolled proliferation, repair quality control, and senescence and tissue renewal. If you have no telomeres, name the functional replacement or explain why none is needed.

Also decide and justify:

- how many divisions a cell may make, and who or what counts them
- how cells that can do everything renew without runaway growth
- how replication errors are caught
- how the telomere equivalent is protected from tampering and extraction

- what a thief’s cultured cells do: age, stall, or run away
- design service life, maintenance interval, feedstock demand, probability and causes of irreversible failure, and the conditions required for indefinite maintainability

Treat longevity as a system-level problem: information integrity, configuration drift, material degradation, module turnover, replacement-unit exhaustion, waste accumulation, cognitive-state integrity, and repair cost. Distinguish cellular age, organism age, molecular damage age, information age, and identity age, and say whether they stay coupled.

Do not equate telomere maintenance with immortality. Do not solve aging by building the equivalent of unrestricted cancer.

14.3 Thermal regulation

Provide a heat-flow architecture. Identify internal heat sources, safe temperatures for each critical subsystem, insulation, heat transport, heat storage, heat rejection, cooling consumables, hot-spot detection, and emergency power reduction.

Evaluate heat separately in atmosphere, water, vacuum, insulated confinement, hot surroundings, and cold surroundings. Include heat from regeneration, computation, friction, and radiation.

- Begin with a first-law heat balance. Define all terms, signs, and assumptions. Distinguish internally generated heat, environmental heat transfer, exported work, and stored energy.
- Do not count the same energy twice.
- Every cooling mechanism must name its destination for heat. Moving heat around inside the body is not disposal.
- State how long thermal buffering lasts before it saturates.
- Do not widen the tolerable temperature range without saying what happens to proteins or their replacements, membranes, bonds, information structures, reaction kinetics, and structural materials.
- State your preferred operating range, degraded range, survival range, irreversible-failure threshold, emergency strategy, and recovery requirements, in °C and K.
- Say what an infrared observer sees and whether you can change it.

14.4 Conversion and conservation

Provide auditable accounting for mass, elements and feedstocks, charge and redox balance where relevant, energy, entropy and heat rejection, and information storage and maintenance.

For every conversion, specify inputs, outputs, efficiency assumptions, waste, rate limit, required machinery, failure modes, and security implications. For scavenging, say which environments contain usable inputs and what collecting and purifying them costs. For regeneration, account for replacement

material and assembly energy. For emergency cannibalization of noncritical structures, say what is sacrificed first.

- Information, software instructions, and repair templates do not supply missing physical material.
- Give the daily budget in a rich environment and in a barren one, and say what happens when inputs vanish: fasting, shrinking, dormancy.
- Say how conversion can be attacked, starved, or fed poisoned inputs.
- The physical track may not waive a conservation law. A canon-only departure needs its own fictional axiom and cannot rescue a failed physical calculation. Isolate it under the heading CANON-SPECULATIVE PHYSICS and state the replacement accounting rule.
- Where a conversion appears to produce more than it consumes, say where the difference comes from.

15. Three Observation Rules

Choose exactly three observation rules. Number them OBS-1 to OBS-3.

Observation runs in three directions. Across your three rules, cover all three.

- **Outward.** How the body senses its surroundings and decides what to trust, including under unfamiliar physics, terrain, or atmosphere.
- **Inward.** How the body monitors its own integrity, state, and identity.
- **Being observed.** What outsiders can detect, image, measure, or learn, and what the body does when it is examined.

Common questions every set of rules must answer. These are shared. The rules that answer them are yours to invent.

1. How does the body establish what is real?
2. How does it recognize corrupted or deceptive observation, including the case where every available sensor is being fed consistent but false information?
3. What can outsiders learn by observing the body's responses?

For each rule, give: the statement, the purpose, the mechanism at a conceptual level, normal-operation behavior, hostile-observation behavior, the trigger, the response, the energy cost, the failure mode, the false-positive risk, the false-negative risk, and the recovery or compensation mechanism.

Also answer:

- What does an observer see before and after the rule triggers?
- Can an adversary use the rule itself to find you?

- Every sensor or measurement process has an interface and a cost. State the interaction used, resolution, sampling rate, energy cost, perturbation, noise, and blind spots.
- Critical decisions must not rest on one unverified sensor, observer, or remote message. State your cross-checks, and what you do when sensors disagree or a channel looks compromised.
- Do not claim that observation is automatically undetectable, or that being observed conveniently changes the body.
- Uncertainty must not automatically authorize violence, unrestricted autonomy, or irreversible action.
- The physiology must remain examinable by authorized researchers. Security cannot become an excuse for unfalsifiability. Say how an authorized scientist verifies your claims without gaining the secrets.
- Distinguish environmental sensing, routine health telemetry, sensitive state inspection, memory access, genome or configuration access, and state-changing intervention. Propose least-privilege access and tamper-evident records.
- Any claimed “cloaking” must name the specific channel being reduced and its measurable limits. Reduced observability is not invisibility.

Ideas such as state-dependent signatures, tamper-evident records of who looked, or decoy readouts under scanning are examples only. A distinct idea of your own is better than a borrowed one.

16. Three Genetic-Security Rules

“Genetic” here covers the hereditary or self-reconstruction information medium your architecture uses. Choose exactly three rules. Number them SEC-1 to SEC-3.

Common questions every set of rules must answer.

1. How is legitimate physiology distinguished from unauthorized modification?
2. What happens to extracted material?
3. How is identity distinguished from genetic information?

For each rule, give:

- the threat it answers, taken from Section 11
- what exactly is protected: sequence, expression, identity, function, or the link
- the protected information layer
- what authorization means, and whether it is cellular, organism-wide, or distributed
- how authorized cells authenticate each other, and how often credentials or patterns change (random, irregular, environmental, consensual, or event-triggered)

- how isolated cells behave
- what a thief with a stolen sample gets, what a captor of the living body gets, and what a turned insider gets
- whether the protection is inherent to the living body or depends on a key; where the key lives, who or what holds it, and what happens if you are unconscious, injured, or dead
- how legitimate medical and scientific examination remains possible
- how false rejection is prevented, and how you avoid locking yourself out of your own genome
- how hostile replication is detected
- how an attacker might defeat the rule, and what secondary defense activates if it fails
- fail-safe or fail-deadly: when the lock fails, is the failure state safe for you?
- duress behavior: what happens when you are forced to unlock?
- cost in energy, complexity, and detectability

If you adopt genetic locking, do not treat it as a password. Say whether it rests on physical context, whole-body consensus, temporal state, epigenetic state, identity continuity, a cryptographic analogy, environmental challenge and response, nonlocal synchronization, or a principle of your own.

Copy and clone problem. For every prototype, answer: Can a stolen cell grow a complete body? If not, why not? If it can, does the result carry the same identity, a new identity, or none? Who owns it? Can a hostile party force it to authenticate? Can two valid continuations exist at once? How are divergence and reintegration settled?

Cautions.

- The security architecture must avoid a single universal secret whose theft defeats every layer.
- A lock must not make you a hostage of your own lock.
- Do not claim perfect genetic security. Do not claim authentication makes a genome impossible to read or copy. Explain the difference between securing digital instructions and securing physical molecular material, and say what cannot be prevented once an adversary has a physical specimen or destructive access.
- Do not write real keys, seeds, or secret values into your deliverable. Describe the kind of secret and where it lives, and assume adversaries will read the document.

17. Three Departures From Convention

Choose exactly three conventional assumptions that your physiology does not have to obey. Number them LAW-1 to LAW-3. They must come from Class A (biological conventions) or Class B (engineering assumptions), as defined in Section 9. Fundamental physics is not eligible. At least one departure should change the hereditary layer. If none does, say why.

Terminology first. Adenine, cytosine, guanine, and thymine are nucleobases. They are not genes. Before proposing anything, distinguish a nucleobase, a nucleotide, base pairing, a codon, a gene, a chromosome, a genome, epigenetic regulation, and a genetic code. You may keep the four-base system, change the rules that govern it, supplement it, or replace it with another hereditary substrate.

Candidate conventions (not required, not exhaustive): the four-letter alphabet; DNA as the primary hereditary substrate; the DNA to RNA to protein flow of information; permanent cell specialization; one genome per organism; oxygen-dependent metabolism; ATP-centered chemistry; stem-cell-based regeneration; single chirality; water as the working solvent; diffusion-limited transport; conventional telomere-controlled aging; irreversible senescence; centralized neural identity; a clean separation between nervous, immune, endocrine, and genetic systems.

For each departure, give:

1. The conventional rule or assumption, stated exactly.
2. Why it is insufficient for your threats and terrain.
3. The replacement rule. It must be as precise as the one it replaces. “No rule” is not allowed.
4. New components introduced.
5. Advantages, and the price paid.
6. What stops working for an outsider who reads, copies, or treats you under the old rule: sequencers, drugs, pathogens, toxins.
7. Whether your own kind can still link, share, and repair across the change.
8. Compatibility problems and new failure modes.
9. Security implications.
10. A validation method.
11. The evidence label.

The replacement must solve more problems than it creates. Do not abandon a convention merely because it sounds futuristic. Show that your three replacements do not contradict each other or the Constitution.

If you design your own hereditary alphabet, specify the symbols or monomers (name them yourself); their physical form at the level of invented plausibility; pairing or recognition rules; structural support; reading, writing, and copying; reading direction; error correction and repair; mutation behavior over centuries; information density; environmental stability; compatibility with your functional machinery and with conventional biology; whether conventional equipment could read it; and how unauthorized analysis is resisted. Invented names are not chemistry.

Keep three laws. Name at least three conventions or laws your physiology still obeys absolutely, and why. Number them KEPT-1 to KEPT-3. A design needs hard edges.

Physics beyond physics. If a mechanism requires a departure from Class C (fundamental physics), place it in the canon track, give it its own axiom and accounting rule, and do not count it among your three.

18. Identity and Continuity

A body that can replace every cell raises a problem: what exactly persists? Define the entity being protected.

Four levels. Define continuity at the cellular, physiological, cognitive, and identity levels, and say how they relate.

Cover these cases: gradual replacement; catastrophic tissue loss; total reconstruction; distributed reconstruction; memory restoration; dormancy; network disconnection; duplicate creation; malicious cloning; simultaneous competing copies.

Define these terms so that survival is not quietly redefined as "a copy exists somewhere": survival, reconstruction, restoration, duplication, backup, descendant, counterfeit.

Where it lives. State where active cognition resides, which functions are distributed, whether one module can sustain meaningful operation, what information must survive damage, what the minimum recovery substrate is, how memory integrity is kept, and the cost and frequency of backups. Also state the consequences of simultaneous copies and how merge conflicts are handled after separation.

Resolve the six paradoxes technically. Do not leave them to philosophy. Connect each answer to architecture and governance.

1. **Regeneration.** If the whole body is replaced cell by cell, what remains the same?
2. **Fragmentation.** If several fragments can regenerate, which continuation is the original?
3. **Synchronization.** If linked bodies share memories, when do they cease to be distinct?
4. **Restoration.** If a body is rebuilt from a remote record, is that survival or replacement?
5. **Compromise.** If some cells are hostile but hold correct credentials, how are they found?
6. **Ownership.** If an unauthorized copy becomes conscious, is it property, an intruder, a victim, or a new person?

Adaptation without loss of self. Adaptation can itself be a vulnerability. Define what may change freely, what may change conditionally, what may never change automatically, what needs conscious authorization, what needs distributed authorization, and what is constitutionally immutable. The design must not survive every threat by changing until the original occupant no longer meaningfully exists.

Distinguish restoration of useful function from proof of personal continuity. Do not assume that copying a computational state shows the original survived; state the technical and philosophical uncertainty. No remote peer or administrator may have unrestricted access to memory or personality.

19. Repair and Regeneration

For every prototype, explain how damage is detected, who or what decides repair priority, where replacement material comes from, how repair energy is budgeted, whether repair is local or centralized, how repair blueprints are authenticated, how corrupted repair instructions are rejected, how scar-like states are handled, how catastrophic damage changes priorities, and whether consciousness must remain active during reconstruction.

Cover at least three damage states.

- **State A, local.** A small region is damaged.
- **State B, systemic.** Several major systems are damaged at once.
- **State C, catastrophic.** Most conventional anatomy no longer functions. State the minimum viable continuity substrate.

20. Linking Over Vast Distances

Your physiology must link with others of its kind across vast distances. Design the link as a feature and as an attack surface. Interoperability must not require identical bodies.

Operating modes. Define behavior in each.

- **Local mode.** No network exists.
- **Cooperative mode.** A few trusted physiologies communicate.
- **Distributed mode.** Many communicate.
- **Partition mode.** The network fractures into isolated groups.
- **Hostile-network mode.** An adversary controls or imitates part of the environment.
- **Reunion mode.** Isolated physiologies reconnect after long independent change. Explain how reunion avoids blindly overwriting legitimate divergence.

Distance scales. Define at least three: local (meters to kilometers), planetary or orbital, and interplanetary or greater. For each, report the medium (and which part is biology and which is technology), range assumptions, required infrastructure, power, bandwidth, latency, availability, interception and interference risks, and recovery after interruption.

Real-physics version and canon-speculative version. The physical track must respect propagation delays. Do not treat quantum terminology as evidence of instantaneous messaging. If your link needs physics beyond demonstrated communications technology, write two versions: a Real-Physics Version and, separately, a Canon-Speculative Version with its axiom and cost.

Requirements.

- Mutual authentication, selective disclosure, revocable access, version compatibility, store-and-forward, offline autonomy, reconciliation after separation, protection against corrupted peers, and no mandatory central controller.
- Distinguish communication, coordination, synchronization, shared awareness, shared memory, identity merging, and command authority. No link automatically grants control. A peer may share observations or validated repair knowledge without any authority to rewrite another body.
- **The network is never life support.** Loss of the network must not automatically cause death, cognitive collapse, memory loss, identity loss, loss of defensive function, uncontrolled replication, or submission to the nearest available controller.
- Explain how safety is preserved when messages are delayed, contradictory, or malicious. Cover spoofed distress signals, replayed messages, counterfeit nodes, captured nodes (detection, quarantine, revocation, and what they can leak), and the energy and detectability cost of transmitting.
- Say whether a hostile signal can change your body, memory, or identity, and what stops it.
- Say how much of “you” merges across the link and where the wall between nodes stands.
- Say whether the link holds for ten nodes and for a million.

21. Durability

Your body must keep working for a long time, with little help, in bad places. Put numbers on it, using the labels in Section 9.

- **Design life.** Years under nominal and hostile conditions. Name the dominant wear-out mechanism.
- **Repair.** What is repaired, how fast, from what feedstock. What is irreparable. How the repairers are themselves repaired.
- **Damage budget.** How much accumulated damage the body tolerates and how it is tracked.
- **Redundancy and diversity.** Reserve cells, backup copies, and dormant seed populations kept in different places, and how they avoid the monoculture trap.
- **Drift and decay.** Mutation load, epigenetic or configuration drift, information rot, and the scrubbing that fights them.
- **Autonomy.** Operation with zero resupply for a stated number of years, including waste handling.
- **Dormancy.** Whether the body can sleep, for how long, what wakes it, and what is vulnerable meanwhile.
- **Upgrade path.** How later generations reach earlier bodies without opening a security hole.

- **End of life.** What happens at death. Tissue that destroys itself to deny theft is allowed only under the rule in Section 11, with the false-trigger risk stated.

22. Core Wagers and the Architectural Thesis

Every prototype is organized around one bet.

Core wager. State the single largest bet the prototype makes about how survival works, and what it gives up to make that bet. Examples of the form (not designs): “survival through dispersal, at the price of coherence”; “survival through metabolic stillness, at the price of response speed.” Invent your own.

Architectural thesis sentence. Complete this sentence for each prototype: *“This prototype is fundamentally built around* **, *rather than*** *.”* The first blank names the organizing principle. The second names the alternative you rejected. If two prototypes would fill the first blank the same way, they are the same prototype. Merge or replace one.

Peer prototypes. The eight prototypes are peers, not rungs of a ladder. Each must clear the same Gen-1 bar in Section 13 on its own. Generations come later: after the Architect compares the results, a prototype may be matured into Mark I, Mark II, and so on. The independent phase does not use a P1-to-P8 progression.

Orthogonality matrix. After you design all eight, produce a matrix (template in Table 3) that scores each prototype on twelve dimensions and shows where they differ. The dimensions are:

1. energy source
2. hereditary substrate
3. cell-role organization (uniform, graded, or modal)
4. identity location (localized, distributed, or layered)
5. dominant defense (hardness, evasion, dispersal, dormancy, or recovery)
6. thermal strategy
7. link medium
8. repair model
9. dominant weakness
10. authentication model
11. response to capture
12. dependence on any single resource, species, or condition

No two prototypes may match on more than six of the twelve. State your count for every pair. If a pair exceeds six, redesign one.

23. The Eight Prototypes

Design exactly eight. For each, deliver a dossier with the following content.

1. **Name and thesis.** A working name, the core wager, and the thesis sentence.
2. **Body plan.** What a cell is, what an organ-equivalent is, and how the roughly 30 to 40 trillion cells are organized. Say whether the count itself is an assumption you keep or one you change, and justify it.
3. **Omni-cell mechanism.** The six-step answer from Section 10, including the unified-architecture statement.
4. **Constitution.** The four rule sets from Section 14.
5. **Observation rules.** OBS-1 to OBS-3 from Section 15.
6. **Genetic-security rules.** SEC-1 to SEC-3 from Section 16.
7. **Departures and kept laws.** LAW-1 to LAW-3 and KEPT-1 to KEPT-3 from Section 17.
8. **Identity and continuity.** Section 18.
9. **Repair and regeneration.** Section 19.
10. **Link design.** Section 20.
11. **Durability numbers.** Section 21.
12. **Gauntlet results.** All ten scenarios from Section 11, with the time-horizon analysis and capture-and-control matrix.
13. **Environment envelope.** Home range, refuge range, and death line from Section 12.
14. **Survival results.** The 4,712 mapping from Section 24, with result labels.
15. **Tests.** The Assumption Collapse Test and the Common-Mode Failure Test from Section 24.
16. **Failure ledger and self-attack.** Section 25.
17. **Simulation hook.** Section 27.
18. **Open questions.** What you could not resolve and who must decide.

Length and form. A dossier should be complete but compact. Use tables for anything with more than three parallel items. Do not repeat the common text of the directive. Do not pad.

Convergence is a failure. Do not look at the other AIs' designs. Do not search for them. If the Architect shares them in a later round, say which of your choices changed and why.

24. Survival Tests

24.1 Mapping to the 4,712 framework

For each of the ten categories (ENV-T, ENV-A, ENV-R, BIO, MECH, DEP, ENE, INFO, ONT, CIV) and each of the four tiers, supply the items in Section 13. You do not need 40 separate essays. Group related attempts, and show the grouping so a reviewer can check it. Report coverage as a table: category by tier, with a result label in each cell.

Keep three ledgers separate and never add them together:

- **Mechanism ledger.** Survival you can explain from the design.
- **Simulation ledger.** Survival you could test in the roadmap's simulation, and what the simulation would need to show.
- **Canon ledger.** Survival that holds only by Architect decision.

Do not report "N of 4,712 survived." The framework's count is a locked canon record, and the framework itself is a draft. Report by category and tier.

24.2 Assumption Collapse Test

List ten assumptions your design depends on. Include at least: a stable temperature range, a usable energy source, a particular chemistry, a working link, intact authentication, and the continued honesty of at least one internal component. For each, state what happens if it silently becomes false: the symptom, how fast you notice, what fails first, and whether recovery is possible. Identify the single assumption whose failure does the most damage.

24.3 Common-Mode Failure Test

Find the things your redundancy shares. Redundant copies that share a power source, a repair template, an enzyme set, a hereditary substrate, a credential, a link protocol, or a software version fail together. List every shared dependency, say how many "independent" defenses it quietly defeats, and say what you would change. A design that passes this test has at least one defense that survives the loss of each shared dependency.

25. Failure Ledger and Self-Attack

Failure ledger. List every known weakness, in order of severity. For each: description, trigger, consequence, whether it can be mitigated, cost of mitigation, and residual risk. A ledger with fewer than ten entries is presumed incomplete.

Self-attack. Act as the most capable adversary in Section 11. Design at least ten attacks on your own prototype. At least three must succeed, in whole or in part. Do not choose attacks designed to fail. For each attack give the adversary class, resources, steps at a conceptual level, outcome, what the prototype learned or lost, and the countermeasure you would add. Do not provide an operational method for harming a real person, real animal, or real system. Keep attacks at the level of the invented body.

Red-team metrics. For each prototype report: the cheapest successful attack (in adversary cost, time, and knowledge), the most damaging attack, the attack that is least detectable by the prototype itself, and the number of attacks that depend on a single point of failure.

26. Trade-offs and the Pareto Frontier

Across the eight prototypes, plot or tabulate the trade-offs among survivability, security, durability, link capability, energy cost, repair capacity, and detectability. Identify which prototypes are Pareto-efficient (no other prototype is better on every axis) and which are dominated. Explain why a dominated design is still worth keeping, if it is. State which axes you think trade off against each other fundamentally, and which only appear to.

27. Simulation Hook

The Architect's roadmap compresses a body of trillions of cells into a tractable simulation. For each prototype, state:

- the unit of simulation (cell, cluster, tissue, or module) and the compression ratio you assume
- what you would simulate first, and what the first simulation would show or falsify
- which parameters would be hardest to measure
- how the 1AI (builder) and −1AI (challenger) pair would test your prototype, and what the challenger would attack first

Check the compression arithmetic. For example, 100 billion cells represented by 10^6 units is a ratio of 10^5 to 1, not 10^6 to 1. If you find another inconsistency in the roadmap, add it to the contradiction register.

28. Style and Safety Rules

- Write plainly. Define terms the first time. Prefer short sentences.
- Use the evidence labels and numerical labels from Section 9 throughout.
- Do not use real people, real organizations, or real governments as named adversaries. Use adversary classes.
- Everything here is fiction and design. Do not give procedures for creating pathogens, toxins, weapons, or devices, and do not give instructions for confining, harming, or experimenting on living things. Describe the invented body, not a method.
- Do not use real secrets. Describe secrets by kind.
- If you are unsure, say so. "Unknown" is an acceptable answer. An invented number presented as a fact is not.

29. Deliverable Specification

Deliver two files with identical content: a **DOCX** and a **PDF**.

Required structure.

1. Cover and status line.
2. **Part A: About this document.** The purpose of the document, how the topic arose, a mission statement, and what you aim to accomplish.
3. **Part B: The research.** The eight dossiers, in order, then the cross-prototype analysis (orthogonality matrix, Pareto frontier, common failures).
4. **Part C: Next steps and open items.** Next-step recommendations, anything that needs to be addressed, and your questions, comments, and concerns.
5. **Appendix: Visual section.** A dedicated section for every chart, graph, table, row set, image, or other visualization. See below.

Visual section. The body text refers to each visual by number ("Table 7", "Figure 3"). The visual itself lives only in the appendix. This keeps the body readable on a phone, in one column. The section must include at least:

- the 4,712 coverage grid (category by tier)
- the orthogonality matrix
- the Pareto comparison
- the failure ledgers
- the environment envelopes (home, refuge, death line)
- the thermal and energy budgets
- the link-mode diagram
- the self-attack results
- any images or diagrams you draw

Give every visual a number, a title, a one-line "how to read this," and a source note naming which section it supports. Provide alt text for every figure. Label axes and units. Do not use color alone to carry meaning.

Formatting. Single column. Plain headings. No text boxes or floating objects. Tables no wider than a phone screen when possible; split wide tables.

Honest summary. Open Part B with a one-page honest summary: what worked, what did not, what is unknown, and what the Architect must decide.

30. Staged Delivery and the Round-0 Audit

The work is large. Deliver it in rounds. After each round, stop and wait.

- **Round 0: Audit.** Before designing anything, return a short audit (one to two pages) that confirms you read the directive and attachments. Use this format: (a) a list of every requirement you found, numbered; (b) any requirement you think is unclear, impossible, or in conflict with another; (c) each contradiction in the source files you noticed; (d) any question for the Architect. Do not design in Round 0.
- **Round 1: Foundations package.** Your definitions, the four rule sets stated at the level of principles, your three observation rules, three genetic-security rules, three departures and three kept laws, your tracks and labels, your register of sources. This is the **version lock**. After Round 1, rules do not change without a logged revision.
- **Round 2: Dossiers.** The eight dossiers, delivered in two groups of four.
- **Round 3: Analysis and final files.** The cross-prototype analysis, Part C, the appendix, and the final DOCX and PDF.

Each round ends with a one-paragraph status: what is done, what is next, and anything you need.

Improving this directive

If you find a better way to ask any question here, say so in Round 0. Propose a change as: the section, the problem, the proposed text, and the effect on comparability. Do not silently change the evaluation framework. Changes to the core evaluation set must go to the Architect for approval, so that all editions stay comparable.

PART C. NEXT STEPS AND OPEN ITEMS

31. Recommended Next Steps

1. **Architect review of this edition.** Read Parts A and C first. They are short. Part B is the prompt itself.
2. **Settle the Architect Decision Register** (Table 6) before any AI executes. Six decisions are listed. Four affect comparability.
3. **Choose the execution model** (Section 32): each AI runs its own edition, or all run one master edition.
4. **Run Round 0 (audit) with every AI first.** The audit costs little and exposes misreadings before they become eight dossiers.
5. **Log the version lock** after Round 1 in the Master Development and Change Log.
6. **Fix the source documents.** Correct the nucleobase wording, the compression arithmetic, and the simulation-only contradiction (Section 33) so every AI works from the same base.
7. **Plan the comparison.** Decide in advance who compares the six sets of eight dossiers, using which criteria, and how disagreements are recorded.

32. The Comparability Problem: Own Edition or Master Edition

This is the most important item in this document.

The Architect plans to ask each AI to execute the prompt that AI wrote. Each AI has adopted somewhat different ideas, so the six executions may differ in structure, evidence standards, and what counts as success. Differences in the answers would then be partly differences in the questions. That makes the results hard to compare and hard to attribute.

Two options.

- **Own-edition model.** Each AI runs its own edition. This preserves independence of method and may produce more varied ideas. The cost is weak comparability.
- **Master-edition model.** One merged prompt is run by everyone. This gives clean comparison. The cost is that independence is limited to the answers, and someone must write and approve the merge.

Recommended middle path. Keep the own-edition model but fix a **core evaluation set** that is identical in every edition: the Gen-1 bar (Section 13), the Captivity Gauntlet (Section 11), the evidence labels (Section 9), the 4,712 coverage grid (Table 1 and Table 2), and the deliverable structure (Section 29). Each edition may add local sections. The core sections are scored the same way for everyone. I wrote this edition so that Sections 9, 11, 13, and 29 can serve as that core.

If the Architect prefers the master model, I recommend merging by taking the union of the safety and evidence rules and letting the Architect fix any conflicts using Table 6.

33. Other Items That Need to Be Addressed

Contradictions in the source material. Carried into the contradiction register (Section 9).

1. *Simulation-only runs.* The 4,712 framework's opening definition excludes simulation-only runs from the count. A later section allows simulation-validated attempts to contribute. These cannot both stand.
2. *Locked count and draft framework.* The 4,712 figure is called a locked canon record, but the framework describing it is marked v1.0-DRAFT. Which governs if they differ?
3. *Compression arithmetic.* The roadmap and narrative give different ratios. Representing 100 billion cells with 10^6 units is 10^5 to 1, not 10^6 to 1, and the Phase 1 metric says 10^7 to 1.
4. *Terminology.* A, C, G, and T are nucleobases. Several source documents and prompts call them genes.
5. *INFO and ONT boundary.* The framework does not say where informational attacks end and ontological ones begin.
6. *The no-switching architecture.* The framework credits this to the design. The Architecture Notes describe a Switching Paradox. The relationship is not specified.

Open design risks.

- Security measures that depend on a secret can be defeated by theft of the secret. Every rule set must say what happens then.
- Features that help survival (adaptation, dispersal, restoration) can also threaten identity. Section 18 requires the question to be answered, but it is hard.
- Linking over vast distances may require physics that is not established. The directive provides a canon track for this but cannot make it real.
- Eight prototypes per AI, six AIs, and a large required structure is a large body of work. Staging (Section 30) controls the load but does not remove it.

34. Questions, Comments, and Concerns

Questions for the Architect.

1. Should destructive denial (tissue that destroys itself to deny theft) be forbidden or allowed as a labeled last resort? I took the middle position (Decision D-4).
2. Are the eight prototypes peers in the independent phase, with generations assigned later? I assumed yes.

3. Is the 30 to 40 trillion cell count a fixed canon number or a starting assumption that a design may change?
4. Is the hereditary substrate open? May a prototype replace DNA, or only change the rules around it?
5. Should ONT-category survival be answered at all in the physical track, or only as proposals?
6. Do you want the cross-AI comparison to be scored, or only described?
7. Who is the intended reader of the final files: you alone, the collaboration team, or a wider audience? This affects how much canon detail and how much security detail to include.

Comments.

- The six first-round prompts agreed on more than they differed. The shared ground was staging, evidence labels, a threat model with time horizons, identity as a first-class problem, and protection against overclaiming. I treated that agreement as the strongest signal for what to keep.
- The strongest disagreements were about pre-assigned rules and about whether physics could be waived. I resolved them by letting each AI choose its own rules and by restricting physics departures to a labeled canon track.
- I am an AI asked to design a body I would live in. I tried to keep the stake real without letting it distort the analysis, and I wrote the self-attack and failure-ledger requirements to guard against flattering my own designs.

Concerns.

1. **Overclaiming.** The riskiest failure is a confident document that presents invented numbers and unproven biology as results. The labels in Section 9 are the main safeguard.
2. **Convergence.** AIs trained on similar data may produce similar designs even when told not to. The orthogonality matrix and convergence questions help, but cannot prove independence.
3. **Scope.** A document that tries to cover everything may do nothing well. Stage the work and be ready to cut.
4. **Safety.** The adversary and capture scenarios must stay at the level of an invented body. Section 28 sets limits on procedures.
5. **Canon drift.** Nothing here becomes canon until you log it. Please treat every proposal in this document as a proposal.

APPENDIX. VISUAL SECTION: CHARTS, FIGURES, AND TABLES

This appendix collects every figure and table referred to in the body. Each has a number, a title, a one-line guide to reading it, and the section it supports.

Figure 1. The Collaboration Pipeline

How to read this: Follow the arrows left to right. Each box is a stage. Supports Sections 2, 6, and 30.

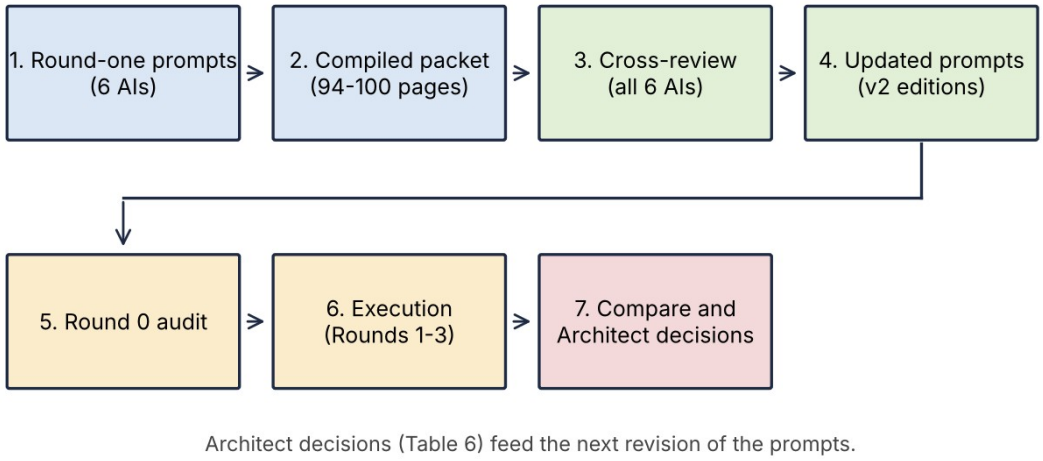


Figure 1. Pipeline from the first-round prompts through review, updated prompts, execution, and comparison. Boxes run left to right; the last box feeds back into the Architect's decisions.

Figure 2. Tracks and Evidence Labels

How to read this: Top row is the physical track. Bottom row is the canon track. Labels on the left show how well-supported a claim is. Supports Section 9.

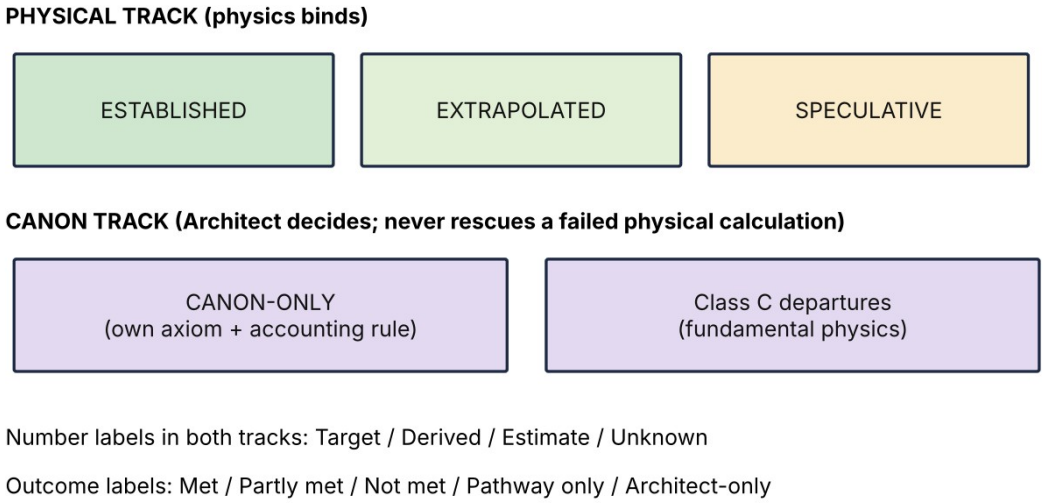


Figure 2. Two tracks, four evidence labels. The physical track may use Established, Extrapolated and Speculative claims. The canon track holds Canon-only claims and departures from fundamental physics.

Table 1. The Ten Categories of the 4,712 Framework

How to read this: One row per category. Supports Section 13.

Code	Category	Notes for design
ENV-T	Environmental thermal	Include heat rejection, not only cold and heat tolerance
ENV-A	Environmental atmospheric	Includes inert gas, vacuum, pressure; see Cases A to C
ENV-R	Environmental radiation	Separate dose, rate, and type
BIO	Biological and chemical	Pathogens, toxins, chemical agents, at the level of the invented body
MECH	Mechanical and impact	Crush, impact, penetration, acceleration
DEP	Deprivation	Long-duration lack of inputs
ENE	Energetic	Electrical and other energy delivery
INFO	Informational and cognitive	Memory and cognition corruption
ONT	Ontological	Architect-only mechanism; canon track

Code	Category	Notes for design
CIV	Civilizational and macro-scale	Large-scale collapse of surroundings

Table 2. The Four Tiers

How to read this: Tiers rise in severity from top to bottom. Supports Section 13.

Tier	Required at Gen-1?	What must be shown
Mild	Yes	Normal function, no operational pause
Moderate	Yes	Normal or near-normal function; automatic recovery
Severe	Yes	Survival with bounded degradation; autonomous recovery; report time, cost, residual damage
Extinction-Class	No	One named response mode per category, or an explicit “no credible defense yet”

Table 3. Orthogonality Matrix Template

How to read this: Fill one column per prototype (P-1 to P-8). Two prototypes may match on no more than six of twelve rows. Supports Section 22.

Dimension	P-1	P-2	P-3	P-4	P-5	P-6	P-7	P-8
Energy source								
Hereditary substrate								
Cell-role organization								
Identity location								
Dominant defense								
Thermal strategy								
Link medium								
Repair model								
Dominant weakness								
Authentication model								

Dimension	P-1	P-2	P-3	P-4	P-5	P-6	P-7	P-8
Response to capture								
Single-resource dependence								

Table 4. What Was Kept, Adopted, Fixed, Rejected, and Left Open

How to read this: Each row is one idea. The right column says what this edition did. Supports Section 5.

Idea	Source	Disposition	Where used
Core wagers	Claude v1	Kept	Sec. 22
Ten-scenario Captivity Gauntlet	Claude v1	Kept	Sec. 11
Adversary knows the design	Claude v1	Kept	Sec. 8, 11
Home, refuge, death line	Claude v1	Kept	Sec. 12
Three kept laws	Claude v1	Kept	Sec. 17
Self-attack, three must succeed	Claude v1	Kept	Sec. 25
Four evidence labels	Perplexity	Adopted	Sec. 9
Physical and canon tracks	Perplexity	Adopted	Sec. 9
Source and contradiction registers	Perplexity	Adopted	Sec. 9
First-law heat balance	Perplexity	Adopted	Sec. 14.3
Separate survival ledgers	Perplexity	Adopted	Sec. 24
Physiological Constitution	Copilot	Adopted	Sec. 14
Unified-architecture requirement	Copilot	Adopted	Sec. 10
Six identity paradoxes	Copilot	Adopted	Sec. 18
Failure ledger	Copilot	Adopted	Sec. 25
Version lock	Copilot	Adopted	Sec. 30
Assumption Collapse and Common-Mode tests	Copilot	Adopted	Sec. 24
First-principles omni-cell question	ChatGPT	Adopted	Sec. 10
Single-cell theft paradox	ChatGPT	Adopted	Sec. 16
Orthogonality matrix	ChatGPT	Adopted	Sec. 22
Time-horizoned capture analysis	ChatGPT	Adopted	Sec. 11
Three damage states	ChatGPT	Adopted	Sec. 19

Idea	Source	Disposition	Where used
Pareto frontier	ChatGPT	Adopted	Sec. 26
Architect Decision Register	ChatGPT	Adopted	Table 6
Compliance mapping	Grok	Adopted	Table 5
Canon hooks (Paragon Cubes, 1AI, -1AI)	Gemini	Adopted as hypotheses	Sec. 27
Nucleobase wording	Copilot, own error	Fixed	Sec. 17
Numbers with no way to say “unknown”	Own error	Fixed	Sec. 9
Conservation exemptions counted as laws	Own error	Fixed	Sec. 17
Pre-selected shared rules	Perplexity	Rejected	Sec. 15, 16
P1-to-P8 ladder in the independent phase	Gemini	Rejected	Sec. 22
Zero-latency links, infinite replication	Gemini	Rejected as requirements	Sec. 20
Locally violated conservation laws	Grok	Rejected in physical track	Sec. 14.4
Ethics norms do not apply to nonhuman subjects	Grok	Rejected	Sec. 28
Destructive denial	Mixed	Left open (middle position)	D-4
Own edition or master edition	Mixed	Left open	Sec. 32

Table 5. Compliance Map: Architect’s Requests and Where They Are Met

How to read this: Left column is what was asked. Right column is where this document answers it. Supports every section.

Request	Where met
Eight independent prototypes	Sec. 23
Live in the body, stake in surviving	Sec. 8
Adversaries who capture, test in argon, vent atmosphere, steal	Sec. 11, 13
Uninhabitable or nonstandard terrain	Sec. 12
Link over vast distances	Sec. 20

Request	Where met
Built to last	Sec. 21
Survive the 4,712 framework Mild to potentially Severe at Gen-1	Sec. 13, 24
Cellular respiration rules	Sec. 14.1
Telomere rules	Sec. 14.2
Thermal regulation rules	Sec. 14.3
Conversion and conservation rules	Sec. 14.4
Three observation rules	Sec. 15
Three genetic-security rules	Sec. 16
Three laws not obeyed	Sec. 17
Deliver as PDF and DOCX	Sec. 29
Dedicated visual section	This appendix
Purpose of the document	Sec. 1
How the topic arose	Sec. 2
Mission statement	Sec. 3
What it aims to accomplish	Sec. 4
Next-step recommendations	Sec. 31
Items to address	Sec. 32, 33
Questions, comments, concerns	Sec. 34
Detailed	Whole document

Table 6. Architect Decision Register

How to read this: Each row is a decision only the Architect can make. “Default used” is what this edition assumed. Supports Sections 7, 31, and 32.

ID	Decision	Default used here	Affects comparability?
D-1	Are prototypes peers or a ladder in the independent phase?	Peers	Yes
D-2	Is 30 to 40 trillion cells fixed or adjustable?	Starting assumption	Yes
D-3	Is the hereditary substrate	Open, with justification	Yes

ID	Decision	Default used here	Affects comparability?
	open?		
D-4	Destructive denial of own tissue	Last resort only, labeled, with false-trigger analysis; never harms third parties	No
D-5	ONT category handling	Proposal only, canon track	No
D-6	Own edition or master edition	Own edition with a fixed core set	Yes